

Analyzing the Impact of Car Features on Price and Profitability

Project description:

In today's world cars are integral part of modern life, as cars provide individuals with the freedom to travel at their own pace and on their own schedule

As I know everyone have their own personal expression representing a car, with a wide range of models and features serve us individual preferences in terms of comfort, style, and performance. Modern cars are equipped with features that enhance comfort and safety making travel more enjoyable and secure.

Cars play a vital role in families (mostly in joint families) daily routines, whether it's transporting children to school, attending social events, or going on family vacations.

The automotive industry has been rapidly evolving over the past few decades, with a growing focus on fuel efficiency, environmental sustainability, and technological innovation. With increasing competition among manufacturers and a changing consumer landscape, it has become more important than ever to understand the factors that drive consumer demand for cars.

In recent years, there has been a growing trend towards electric and hybrid vehicles and increased interest in alternative fuel sources such as hydrogen and natural gas. At the same time,

traditional gasoline-powered cars remain dominant in the market, with varying fuel types and grades available to consumers.

In this project it aims to address is to analyse how various features of cars (e.g., brand, model, engine type, colour, safety features, technology, fuel efficiency) influence their market price and profitability for manufacturers and dealerships. This analysis will help business partner/stakeholders in the automotive industry make informed decisions about car pricing strategies, design improvements, and feature prioritization to maximize profitability.

For the given dataset, as a Data Analyst, the client has asked How can a car manufacturer optimize pricing and product development decisions to maximize profitability while meeting consumer demand?

This problem could be approached by analyzing the relationship between a car's features, market category, and pricing, and identifying which features and categories are most popular among consumers and most profitable for the manufacturer. By using data analysis techniques such as regression analysis and market segmentation, the manufacturer could develop a pricing strategy that balances consumer demand with profitability, and identify which product features to focus on in future product development efforts. This could help the manufacturer improve its competitiveness in the market and increase its profitability over time.

As a Data Analyst, always when the raw data has been given it's a pre requisite to clean the data according to dataset it includes variety of processes either it could remove null cells, blank cells, duplicates, handling missing values using descriptive statistics and outliers if detected

The dataset contains information on various car models and their specifications, and is titled "Car Features and MSRP". It was collected and made available on Kaggle by Cooper Union; a private college located in New York City.

Here is a brief overview of the dataset:

- **Number of observations:** 11,159
- **Number of variables:** 16
- **File type:** CSV (Comma Separated Values)

The variables in the dataset are:

- **Make:** the make or brand of the car
- **Model:** the specific model of the car
- **Year:** the year the car was released
- **Engine Fuel Type:** the type of fuel used by the car (gasoline, diesel, etc.)
- **Engine HP:** the horsepower of the car's engine
- **Engine Cylinders:** the number of cylinders in the car's engine
- **Transmission Type:** the type of transmission (automatic or manual)
- **Driven_Wheels:** the type of wheels driven by the car (front, rear, all)
- **Number of Doors:** the number of doors the car has
- **Market Category:** the market category the car belongs to (Luxury, Performance, etc.)
- **Vehicle Size:** the size of the car
- **Vehicle Style:** the style of the car (Sedan, Coupe, etc.)
- **Highway MPG:** the estimated miles per gallon the car gets on the highway
- **City MPG:** the estimated miles per gallon the car gets in the city

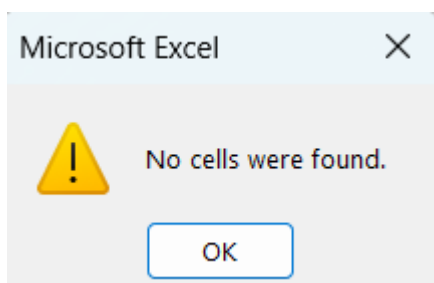
- **Popularity:** a ranking of the popularity of the car (based on the number of times it has been viewed on Edmunds.com)
- **MSRP:** the manufacturer's suggested retail price of the car

This dataset could be useful for a variety of data analysis tasks, such as:

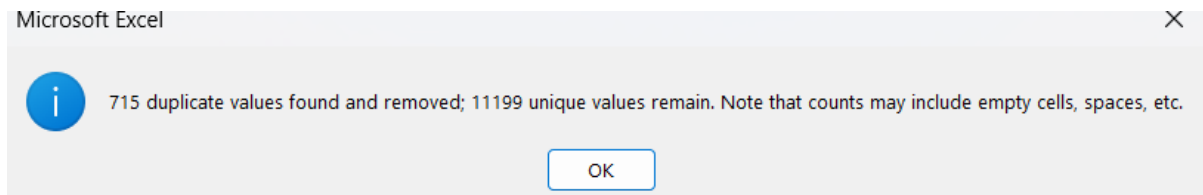
- Exploring trends in car features and pricing over time
- Comparing the fuel efficiency of different types of cars
- Investigating the relationship between a car's features and its popularity
- Predicting the price of a car based on its features and market category

I had never known there were various factors affect the Impact of Car Features on Price and Profitability as I was analysing the data I had assumptions as how one factor affects other, how defect in one trait could drop a deal, how important is to provide detailed information of everything so that it could have a genuine review and finalising one is actually a task,while trying to complete the task either to derive information from data or representing graphically how providing different insights, trends and patterns might profit both sellers & dealers,

After I got the data, I've started the cleaning process started with count of blank cells then deleted the blank cells



Removed duplicates



if missing values found dealt with descriptive statistics

Also done the EDA(exploratory data analysis)mostly with descriptive visualization tools to represent data graphically also to present dashboards

Approach:

The approach for analyzing the impact of car features on price and profitability, incorporated multiple steps, tools, and methods. This structured approach will allow us to derive insights from the given data

- Ensuring data quality by removing inconsistencies, handling missing values, and standardizing data formats whether it could be Computing basic summary statistics (mean, median, standard deviation)
- Using charts, scatter plots, and histograms to visualize the relationships between different features (e.g., engine type, transmission type, fuel efficiency, car brands) and car prices add on elements either it could be trendlines to represent relationship /axis and so on...
- performed Correlation Analysis to identify which features have a strong relationship with car prices.
- Regression Analysis to quantify the impact of various car features on the car's price. The model will include features as independent variables and car price as the dependent variable.

Given tasks based on the business problem so it would require advanced Excel skills and knowledge of data analysis

techniques such as regression analysis, pivot tables, and building an interactive dashboard, a data analyst could provide valuable insights to a car manufacturer and help them optimize their pricing and product development decisions to maximize profitability while meeting consumer demand.

From a data analyst perspective after doing some beginner level projects and intermediate level projects we've come to a point where every query is precisely understood and can exactly know how to retrieve information, trends, patterns, what analysis to be done etc...so that it eventually helps businesses to acquire what they are searching for

Tech-Stack Used:

Microsoft Excel: For Data cleaning process, various Data analytical tools & skills to provide valuable insights

Microsoft Word: To Document everything what I've done during the process and how I gained so many insights dealt with advanced excel skills either it could be detecting a problem , having variety approaches , choosing what best works for a particular task , extracting and providing relative information required

Canva: To present project presentations serves as an essential communication tool that helps present complex information in an understandable way

Insights

Tasks: Analysis

Insight Required: How does the popularity of a car model vary across different market categories?

Task 1.A: Create a pivot table that shows the number of car models in each market category and their corresponding popularity scores.

Ans: Here in this task first after data cleaning process here we need to create a pivot table to know Popularity of a Car Model Vary Across Different Market Categories

So, I've taken out market category in to rows, For the values I've taken count of car models and average of popularity and created a pivot table

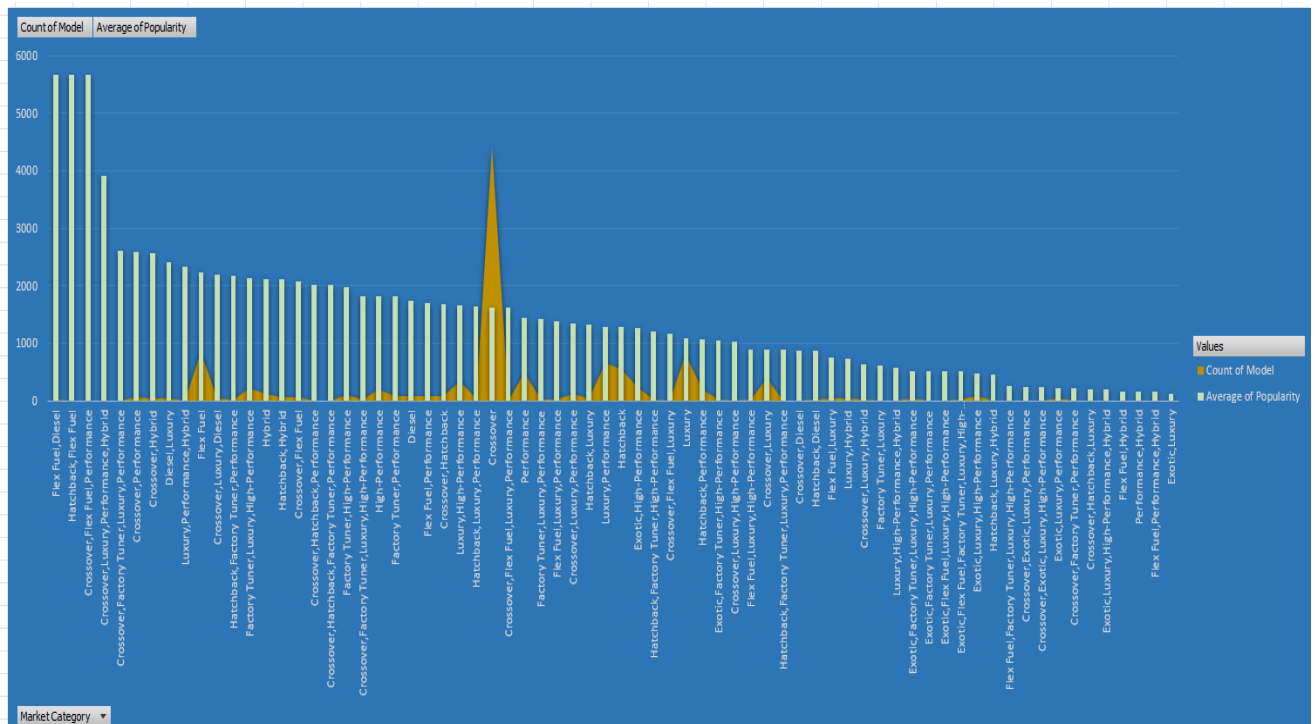
Result:

Popularity of a Car Model Vary Across Different Market Categories		
Row Labels	Count of Model	Average of Popularity
Flex Fuel,Diesel	16	5657
Hatchback, Flex Fuel	7	5657
Crossover, Flex Fuel, Performance	6	5657
Crossover, Luxury, Performance, Hybrid	2	3916
Crossover, Factory Tuner, Luxury, Performance	5	2607.4
Crossover, Performance	69	2585.956522
Crossover, Hybrid	42	2563.380952
Diesel, Luxury	47	2416.106383
Luxury, Performance, Hybrid	11	2333.181818
Flex Fuel	855	2225.71345
Crossover, Luxury, Diesel	33	2195.848485
Hatchback, Factory Tuner, Performance	21	2173.714286
Factory Tuner, Luxury, High-Performance	215	2133.367442
Hybrid	121	2116.586777
Hatchback, Hybrid	64	2111.15625
Crossover, Flex Fuel	64	2073.75
Crossover, Hatchback, Performance	6	2009
Crossover, Hatchback, Factory Tuner, Performance	6	2009
Factory Tuner, High-Performance	104	1966.442308
Crossover, Factory Tuner, Luxury, High-Performance	26	1823.461538
High-Performance	198	1823.378788
Factory Tuner, Performance	81	1818.049383
Diesel	84	1730.904762
Flex Fuel, Performance	81	1702.358025
Crossover, Hatchback	72	1675.694444
Luxury, High-Performance	334	1668.017964
Hatchback, Luxury, Performance	36	1632.25
Crossover	4430	1629.943792
Crossover, Flex Fuel, Luxury, Performance	6	1624
Performance	503	1443.234592
Factory Tuner, Luxury, Performance	31	1413.419355
Flex Fuel, Luxury, Performance	28	1380.071429
Crossover, Luxury, Performance	112	1349.089286
Hatchback, Luxury	45	1323.133333
Luxury, Performance	659	1293.062215

Task 1.B: Create a combo chart that visualizes the relationship between market category and popularity.

Ans: To visualize the relationship between market category and popularity I've used combination chart to understand relationship more efficiently

Result:



- ‘Flex Fuel, Diesel’ Hatchback, Flex, fuel’, and Crossover, Flex Fuel, Performance models are very popular
- Maximum 4430 models belong to crossover category in the market.
- Exotic, Luxury Models are the very low in popularity
- “Flex Fuel,Factory Tuner,Luxury,High-Performance”, “Crossover,Exotic,Luxury,Performance”, “Crossover,Exotic,Luxury,High-Performance”,

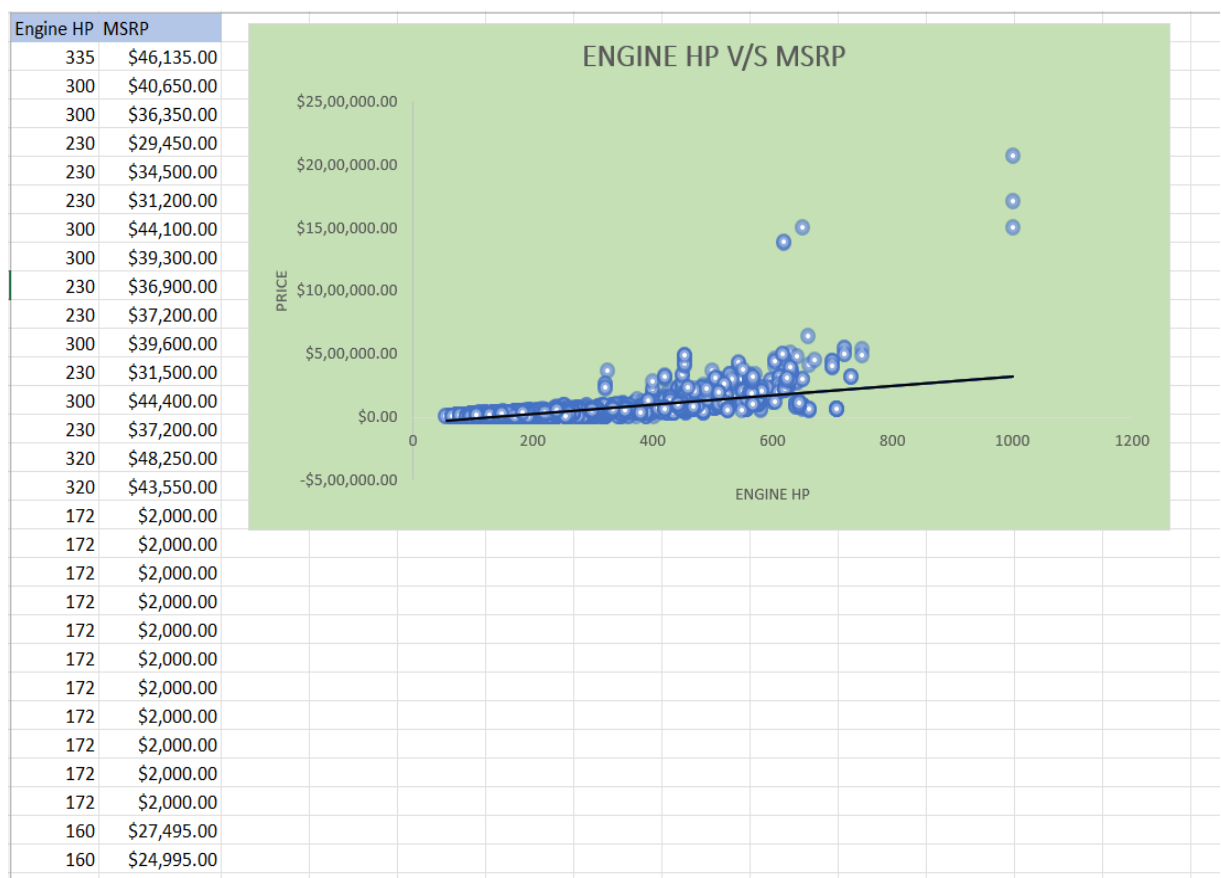
“Exotic,Luxury,High-Performance,Hybrid”,

“Performance,Hybrid” category might belong to luxury models and have single model available in Market.

Insight Required: What is the relationship between a car's engine power and its price?

TASK 2: Create a scatter chart that plots engine power on the x-axis and price on the y-axis. Add a trendline to the chart to visualize the relationship between these variables

Ans: Took out the column's engine power and price and inserted scatter chart on x-axis engine power ,on y-axis price to visualize their relationship



In the scatter plot graph trendline moving upward so its showing the positive relationship between price and engine power, it means as the engine power increasing price is also increasing.

Insight Required: Which car features are most important in determining a car's price?

Task 3: Use regression analysis to identify the variables that have the strongest relationship with a car's price. Then create a bar chart that shows the coefficient values for each variable to visualize their relative importance.

Ans: Here for this task, I've taken out the columns which have the strongest relationship with a car's price they are

- Engine HP
- Engine Cylinders
- Number of Doors
- highway MPG
- city mpg
- Popularity
- MSRP

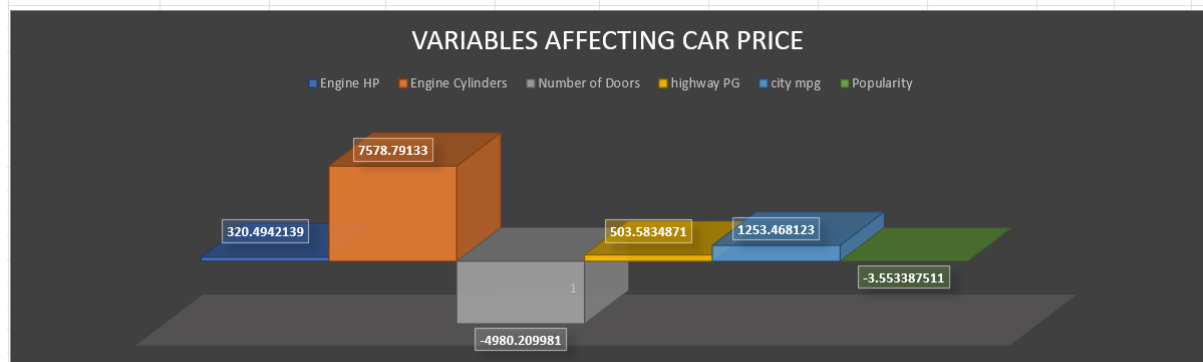
Here after selecting this data, I clicked on data tab then data analysis then we've got list of options their we need to select regression then we get to fill the y variable which we can give a dependent variable which is (MSRP) and independent variable which are all of the above we can either select for each one or all at one's, I selected all at once and then ok then we get it as

Result:

Engine HP Engine Cylinders Number of Door highway MPG city mpg Popularity MSRP						
335	6	2	26	19	3916	\$46,135.00
300	6	2	28	19	3916	\$40,650.00
300	6	2	28	20	3916	\$36,350.00
230	6	2	28	18	3916	\$29,450.00
230	6	2	28	18	3916	\$34,500.00
230	6	2	28	18	3916	\$31,200.00
300	6	2	26	17	3916	\$44,100.00
300	6	2	28	20	3916	\$39,300.00
230	6	2	28	18	3916	\$36,900.00
230	6	2	27	18	3916	\$37,200.00
300	6	2	28	20	3916	\$39,600.00
230	6	2	28	19	3916	\$31,500.00
300	6	2	28	19	3916	\$44,400.00
230	6	2	28	19	3916	\$37,200.00
320	6	2	25	18	3916	\$48,250.00
320	6	2	28	20	3916	\$43,550.00
172	6	4	24	17	3105	\$2,000.00
172	6	4	20	16	3105	\$2,000.00
172	6	4	21	16	3105	\$2,000.00
172	6	4	24	17	3105	\$2,000.00
172	6	4	20	16	3105	\$2,000.00
172	6	4	21	16	3105	\$2,000.00
172	6	4	21	16	3105	\$2,000.00
172	6	4	22	16	3105	\$2,000.00
172	6	4	22	17	3105	\$2,000.00
172	6	4	22	16	3105	\$2,000.00
172	6	4	21	16	3105	\$2,000.00

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.683387437							
R Square	0.46701839							
Adjusted R Square	0.466730032							
Standard Error	45078.99614							
Observations	11097							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	6	1.9747E+13	3.29117E+12	1619.57869	0			
Residual	11090	2.25362E+13	2032115893					
Total	11096	4.22832E+13						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-97167.8727	3898.078612	-24.9271199	1.714E-133	-104808.8	-89526.9451	-104808.8	-89526.9451
Engine HP	320.4942139	6.3774589	50.25421863	0	307.9932598	332.995168	307.9932598	332.995168
Engine Cylinders	7578.79133	461.2602827	16.43061762	5.8865E-60	6674.639109	8482.94355	6674.639109	8482.94355
Number of Doors	-4980.20998	496.4047724	-10.0325586	1.382E-23	-5953.25165	-4007.16831	-5953.25165	-4007.16831
highway PG	503.5834871	109.2773107	4.608307836	4.1049E-06	289.3805157	717.7864585	289.3805157	717.7864585
city mpg	1253.468123	125.6629389	9.974843293	2.4629E-23	1007.146405	1499.789841	1007.146405	1499.789841
Popularity	-3.55338751	0.297352947	-11.9500666	1.0299E-32	-4.13625219	-2.97052283	-4.13625219	-2.97052283

And also mentioned to created a bar chart to that shows the coefficient values for each variable to visualize their relative importance so it looks like



In the graph Engine cylinder having the most positive relation whereas Number of Doors having the most negative relation with Car Price, it means number of doors increasing then car price decreasing.

TASK 4: How does the average price of a car vary across different manufacturers?

Task 4.A: Create a pivot table that shows the average price of cars for each manufacturer.

Ans: Here, I've taken out car brand in to row and price column into average value as pivot table to show the average price of cars for each manufacturer

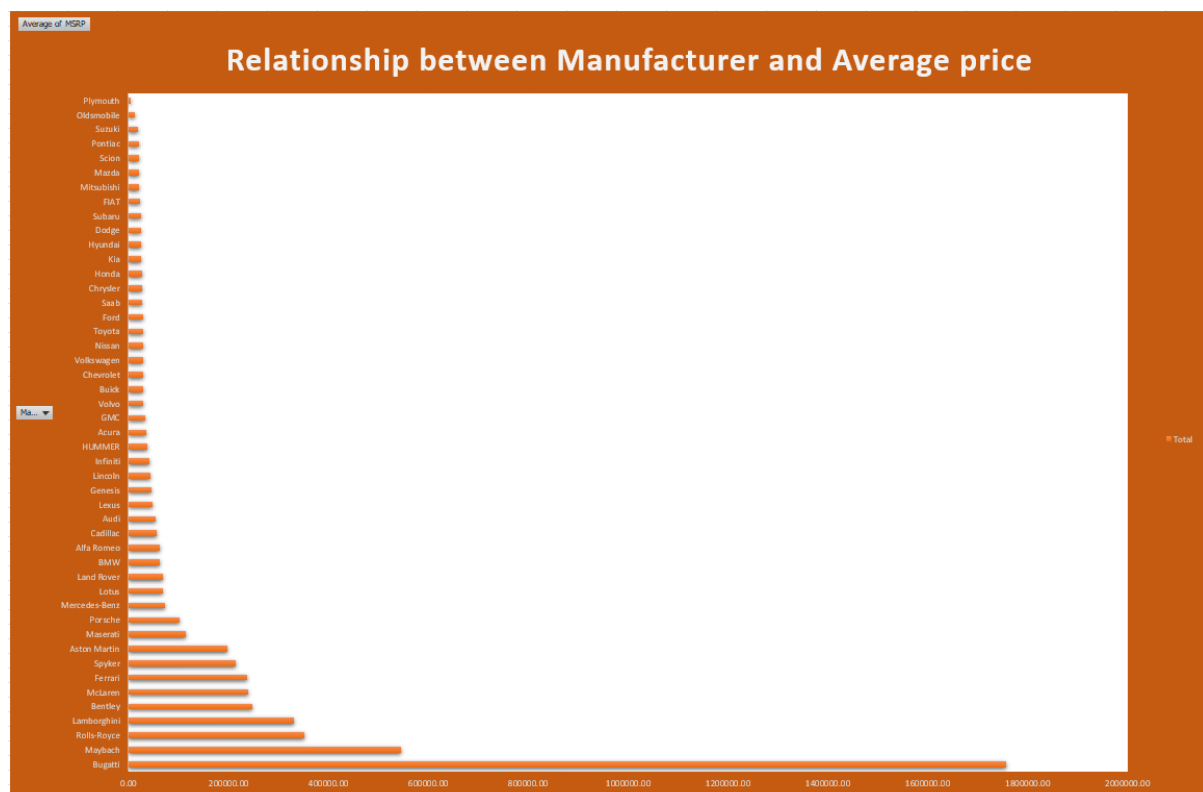
Result:

Average Price of a Car Vary Across Different Manufacturers

Manufact	Average of MSRP				
Bugatti	1757223.67				
Maybach	546221.88				
Rolls-Royce	351130.65				
Lamborghini	331567.31				
Bentley	247169.32				
McLaren	239805.00				
Ferrari	237383.82				
Spyker	214990.00				
Aston Martin	198123.46				
Maserati	113684.49				
Porsche	101622.40				
Mercedes-Be	72135.03				
Lotus	68377.14				
Land Rover	68067.09				
BMW	62162.56				
Alfa Romeo	61600.00				
Cadillac	56368.27				
Audi	54574.12				
Lexus	47549.07				
Genesis	46616.67				
Lincoln	43560.01				
Infiniti	42640.27				
HUMMER	36464.41				
Acura	35087.49				
GMC	32444.09				
Volvo	29724.68				
Buick	29034.19				
Chevrolet	29000.22				
Volkswagen	28947.37				
Nissan	28856.42				
Toyota	28758.77				
Ford	28522.86				
Saab	27879.81				
Chrysler	26722.96				
Honda	26608.88				
Kia	25318.75				
Hyundai	24926.26				
Dodge	24857.05				
Subaru	24240.67				
FIAT	22206.02				
Mitsubishi	21316.35				
Mazda	20106.56				
Scion	19932.50				
Pontiac	19800.04				
Suzuki	18021.05				
Oldsmobile	12843.80				
Plymouth	3296.87				
Grand Tota	41901.12				

Task 4.B: Create a bar chart or a horizontal stacked bar chart that visualizes the relationship between manufacturer and average price.

Ans: created a horizontal stacked bar chart to visualize the relationship between manufacturer and average price.



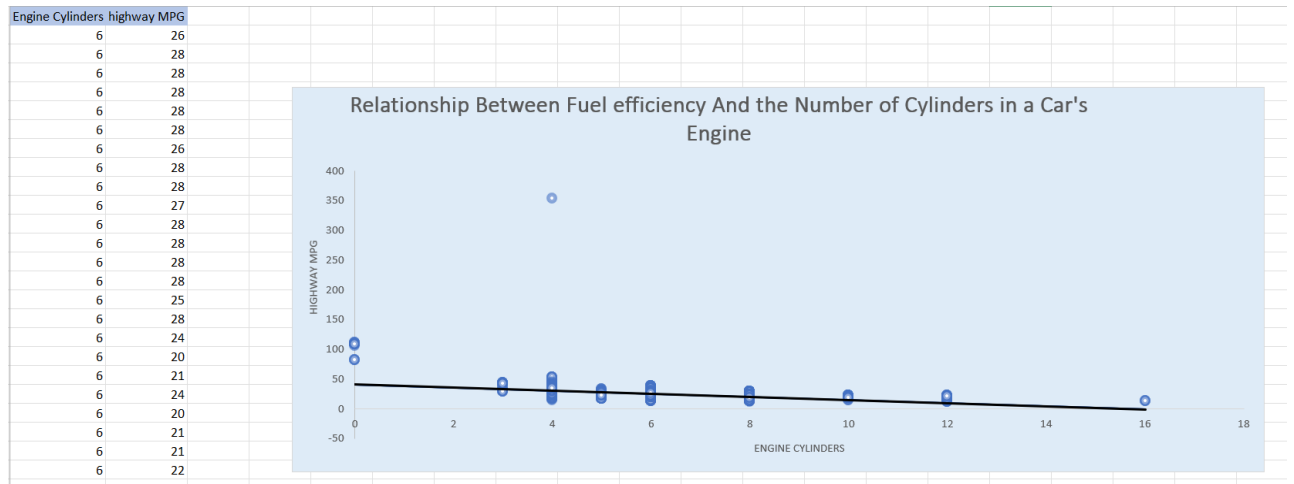
This analysis shows the average price difference of each car, some company like Bugatti having the very high price followed by Maybach, Rolls-Royce and Lamborghini, And the less prices car like Plymouth, Oldsmobile, Suzuki.

Task 5: What is the relationship between fuel efficiency and the number of cylinders in a car's engine?

Task 5.A: Create a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis. Then create a trendline on the scatter plot to visually estimate the slope of the relationship and assess its significance.

Ans: For this task I've taken out engine cylinder and highway mpg column created a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis added a trendline plot to visually estimate the slope of the relationship and assess its significance

Result:



In the graph the trendline going downwards which is showing negative slope result is declining of Highway milage as per engine cylinder increase

Task 5.B: Calculate the correlation coefficient between the number of cylinders and highway MPG to quantify the strength and direction of the relationship

Ans: To calculate correlation I've used CORREL (array1, array2)

Correlation Coefficient

-0.6147	
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The correlation between Engine cylinder and Highway mpg is negative, its clearly showing that as the number of cylinders increase in car the Highway milage will also decrease

Building the Dashboard:

Task 1: How does the distribution of car prices vary by brand and body style?

Ans: I've selected make, vehicle style & MSRP column and inserted to pivot table

In value field I've changed it to sum of MSRP and got the pivot table with the distribution of car prices vary by brand and body style and with the pivot table I've also used slicer with same columns, Slicers are used to provide an intuitive and interactive way to filter data, especially when working with PivotTables and Pivot Charts. It provides a visual representation of the filtering process, allowing users to quickly and easily segment data based on different criteria, also using the stacked column chart represented data graphically to understand how car prices vary by brand & body style

Result: [DASHBOARD TASK-1.xlsx](#)

The brand Chevrolet have the maximum number of cars available in different body style & Bugatti having the single variant of body style

Task 2: Which car brands have the highest and lowest average MSRPs, and how does this vary by body style?

Ans: I've selected Make, Vehicle style & MSRP column and inserted to pivot table as well as slicers to quickly and easily segment data based on different criteria

In value field I've changed it to average of MSRP to know which car brands have the highest and lowest average MSRPs, and how does this vary by body style represented graphically with clustered column chart

Result: [TASK-2 DASHBOARD.xlsx](#)

Bugatti brand have the highest average price in Coupe style among all the brands & Plymouth brand having the lowest average price compare to all other brands.

Task 3: How do the different feature such as transmission type affect the MSRP, and how does this vary by body style?

Ans: I've selected MSRP, Vehicle style & Transmission type columns and inserted to pivot table and also created slicers to quickly and easily segment data based on different criteria

Used average of MSRP for value field settings & inserted a Scatter plot chart to visualize the relationship between MSRP and transmission type, with different symbols for each body style

Result: [TASK-3 DASHBOARD.xlsh](#)

From the graph we can see that the automatic transmission types having the higher average MSRP compare to the manual transmission.

Automated manual transmission only available in the luxury brands like Bugatti, Maybach, Lamborghini, Ferrari, Bentley and also average MSRP is very high.

Task 4: How does the fuel efficiency of cars vary across different body styles and model years?

Ans: I've selected year, vehicle style & highway MPG and inserted pivot table & slicers too to quickly and easily segment data based on different criteria

Selected pivot table and represented graphically using Line chart to show the trend of fuel efficiency (MPG) over time for each body style.

Result: [TASK-4 DASHBOARD.xlsx](#)

By using the line chart analysis, it is observed that the “4dr Hatchback” body style have the highest milage next highest “2dr Hatchback” followed by “sedan” body style having the maximum milage

Fuel efficiency have significantly improved in 4dr hatchback during time and have very good milage.

Task 5: How does the car's horsepower, MPG, and price vary across different Brands?

Ans: selected make column, engine HP, highway MPG, MSRP and inserted it to pivot table, used average in value field settings and only made slicer for car brands

And also, with that used Bubble chart to visualize the relationship between horsepower, MPG, and price across different car brands

Result: [TASK-5 DASHBOARD.xlsx](#)

Bugatti may have higher average horsepower compared

It indicates higher horsepower cars tend to be more expensive.

In summary, through this project I’ve gained an experience regarding usage of wide range of advanced Excel skills and various Data analytical techniques such as regression analysis, pivot tables, and building an interactive dashboard which lead me to deal with complex problems at ease

